

Meghdoot AI- A Cloud Native Weather Nowcasting Framework Using INSAT 3DS/3DR Imagery

Abstract

Accurate short-term weather forecasting, commonly known as nowcasting, is critical for disaster management, agriculture, aviation, and public safety. Traditional numerical weather prediction models often require significant computational resources and may struggle to capture rapidly evolving atmospheric phenomena at high temporal resolution. Recent advances in deep learning and generative artificial intelligence have introduced new opportunities for satellite-based weather forecasting through data-driven approaches.

This paper presents Meghdoot-AI, a cloud-native weather nowcasting framework that leverages Latent Diffusion Models (LDMs) for forecasting future satellite observations from historical geostationary satellite imagery. The proposed system utilizes INSAT-3DR/3DS satellite data obtained from the MOSDAC platform, focusing on Thermal Infrared (TIR1) and Water Vapor (WV) channels to capture atmospheric dynamics and cloud evolution patterns. Approximately 300 GB of satellite data were automatically acquired, preprocessed, and transformed into temporal sequences suitable for deep learning-based forecasting. A Variational Autoencoder (VAE) was employed to compress high-dimensional satellite imagery into a compact latent space, where the diffusion process learns atmospheric evolution through iterative denoising.

To evaluate forecasting performance, the latent diffusion framework was compared against a ConvLSTM baseline using standard image quality and meteorological evaluation metrics, including Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), Root Mean Square Error (RMSE), and Critical Success Index (CSI). Experimental results demonstrate that the diffusion model achieves competitive forecasting performance, obtaining an SSIM of 0.726, PSNR of 28.95 dB, and CSI values of 0.875 and 0.786 at medium event thresholds. Visual analysis further indicates that the diffusion model preserves cloud morphology, atmospheric textures, and large-scale weather structures with high realism.

The complete system was deployed on Google Cloud Platform using Docker, Cloud Run, Cloud Build, Cloud Scheduler, and automated CI/CD pipelines, enabling scalable and production-ready weather forecasting. The results demonstrate that latent diffusion models provide a promising alternative to conventional recurrent architectures for satellite nowcasting, offering improved atmospheric realism, efficient computation, and seamless cloud deployment for real-time forecasting applications.

Introduction

Atmospheric nowcasting plays a critical role in modern disaster management, aviation safety, agriculture, hydrological monitoring, and climate-sensitive infrastructure planning. The increasing frequency of extreme weather events, including flash floods, cyclones, cloudbursts, and severe thunderstorms, has significantly

amplified the need for accurate short-term weather forecasting systems capable of providing high-resolution predictions with minimal latency. Traditional forecasting systems based on Numerical Weather Prediction (NWP) models have achieved remarkable scientific progress over the past decades; however, these approaches remain computationally intensive and often struggle to generate highly localized short-term forecasts in real time. Their dependence on large-scale physical simulations, complex atmospheric equations, and expensive computational resources limits their operational efficiency for rapid atmospheric nowcasting applications.

In recent years, deep learning and generative artificial intelligence have emerged as promising alternatives for atmospheric prediction tasks. Advances in convolutional neural networks (CNNs), recurrent architectures such as ConvLSTM, and transformer-based forecasting systems have demonstrated substantial improvements in spatiotemporal modeling capabilities. These methods have shown potential in capturing cloud dynamics, precipitation patterns, and atmospheric evolution directly from satellite imagery. However, many existing approaches continue to suffer from several critical limitations, including blurry predictions, weak preservation of atmospheric structures, loss of fine-grained cloud textures, and high computational overhead caused by direct pixel-space forecasting on high-resolution imagery.

To address these limitations, this research introduces Meghdoot-AI, a cloud-native atmospheric nowcasting framework designed using latent diffusion modeling and geospatial artificial intelligence techniques for forecasting atmospheric evolution using INSAT-3DR and INSAT-3DS satellite imagery. The proposed system integrates modern generative AI methodologies with scalable cloud computing infrastructure to build a robust, high-resolution, operational forecasting pipeline capable of supporting real-time atmospheric intelligence applications over the Indian subcontinent.

The primary objective of Meghdoot-AI is to develop a scalable atmospheric forecasting system capable of generating realistic short-term cloud evolution predictions while minimizing computational complexity through latent-space modeling. Instead of directly forecasting full-resolution satellite imagery, the proposed framework utilizes a Variational Autoencoder (VAE) to compress atmospheric frames into compact latent representations. These latent tensors preserve essential atmospheric structures such as cloud morphology, thermal gradients, moisture patterns, and storm evolution characteristics while significantly reducing memory and computational requirements. A Latent Diffusion Model (LDM) built upon a U-Net architecture is then trained to learn atmospheric evolution dynamics within latent space using iterative denoising processes.

The adoption of latent diffusion forecasting provides several important advantages over conventional forecasting architectures. Diffusion models have recently demonstrated state-of-the-art performance in image synthesis, video generation, and generative modeling tasks due to their ability to iteratively reconstruct realistic outputs while preserving high-frequency textures and structural coherence. In

the context of atmospheric forecasting, these properties become highly valuable because cloud systems evolve through highly nonlinear spatial and temporal interactions. Unlike regression-based forecasting systems that often produce averaged or blurry outputs, diffusion models enable sharper atmospheric predictions with improved preservation of cloud boundaries and mesoscale structures.

The proposed framework utilizes indigenous Indian satellite systems, specifically INSAT-3DR and INSAT-3DS, accessed through the MOSDAC platform. INSAT satellites provide continuous geostationary observations of the Indian subcontinent and surrounding oceanic regions, making them highly suitable for operational atmospheric monitoring and weather intelligence systems. More than 300 GB of Level-1C satellite imagery was acquired using automated MOSDAC ingestion pipelines developed in Python and executed on Google Colab environments. Although INSAT provides multiple spectral channels, the current implementation focuses on two key atmospheric channels: Thermal Infrared 1 (TIR1) and Water Vapor (WV). These channels were selected due to their strong representation of atmospheric thermal distributions, moisture transport, cloud-top temperature characteristics, and large-scale cloud dynamics essential for nowcasting applications.

A major contribution of Meghdoot-AI lies in its cloud-native architecture and scalable deployment strategy. The complete forecasting pipeline was designed using distributed cloud infrastructure integrating Google Cloud Storage (GCS), Dockerized GPU environments, FastAPI-based inference services, Streamlit visualization dashboards, and automated scheduling systems. Data preprocessing and ingestion pipelines were executed using Google Colab and synchronized with cloud storage buckets to enable scalable distributed training workflows. The use of Docker-based containerization ensured reproducibility across development and cloud deployment environments while enabling GPU acceleration using NVIDIA L4 GPU infrastructure.

The preprocessing pipeline developed for this framework includes automated HDF5 parsing, geospatial reprojection, atmospheric channel extraction, normalization, Indian subcontinent cropping, and tensor generation using Rasterio, GDAL, xarray, h5py, and PyTorch. Processed atmospheric tensors were subsequently compressed into latent representations through VAE-based encoding. The trained diffusion architecture learned temporal atmospheric evolution patterns by observing historical latent sequences and generating future latent predictions through iterative denoising operations. The generated latent forecasts were decoded back into visible atmospheric imagery for qualitative and quantitative evaluation.

Experimental validation demonstrated stable convergence behavior during VAE and diffusion training. The VAE achieved strong reconstruction quality with SSIM values of 0.8800, PSNR values of 35.57, and low RMSE, indicating successful preservation of atmospheric structures within latent space. Comparative analysis was also conducted between the proposed diffusion-based forecasting framework and ConvLSTM baseline architectures using SSIM, PSNR, RMSE, and Critical

Success Index (CSI) metrics. Results indicated that while ConvLSTM achieved competitive pixel-level reconstruction metrics, the latent diffusion framework demonstrated superior preservation of cloud morphology, atmospheric continuity, and realistic cloud evolution patterns in qualitative evaluations.

Another significant contribution of this work is the integration of MLOps and cloud deployment workflows into atmospheric forecasting research. Experiment monitoring was implemented using Weights & Biases (W&B), enabling real-time tracking of training metrics, reconstruction quality, diffusion loss evolution, and generated atmospheric outputs. Automated cloud deployment pipelines were developed using Google Cloud Build, Docker Compose orchestration, and cloud scheduling services to support operational scalability and future real-time deployment scenarios.

The Meghdoot-AI framework demonstrates how generative AI, latent diffusion modeling, geospatial intelligence, and scalable cloud infrastructure can be integrated into a unified atmospheric forecasting architecture. The proposed system not only establishes a foundation for operational real-time atmospheric nowcasting over India but also contributes toward the growing intersection of geospatial AI, climate intelligence, and cloud-native machine learning systems.

The remainder of this paper is organized as follows. Section 2 reviews related work in atmospheric nowcasting, diffusion models, and deep learning-based forecasting systems. Section 3 discusses dataset acquisition and cloud-native data engineering workflows. Section 4 presents the proposed system architecture and operational pipeline. Section 5 describes the geospatial preprocessing framework, while Sections 6 and 7 discuss the Variational Autoencoder and latent diffusion forecasting architecture, respectively. Section 8 presents experimental evaluations and comparative results against ConvLSTM baselines. Finally, Sections 9 through 12 discuss deployment infrastructure, challenges, future directions, and conclusions of the proposed research.

Literature Review

Accurate weather nowcasting has long been a challenging problem due to the highly nonlinear and dynamic nature of atmospheric processes. Traditional Numerical Weather Prediction (NWP) models rely on physical equations governing atmospheric motion and thermodynamics. Although these models provide reliable medium- and long-range forecasts, they are computationally expensive and often struggle to capture rapidly evolving weather phenomena at fine spatial and temporal resolutions. Consequently, researchers have increasingly explored machine learning and deep learning techniques to improve short-term weather forecasting using observational data.

One of the earliest breakthroughs in deep learning-based weather forecasting was the introduction of Convolutional Long Short-Term Memory (ConvLSTM) networks by Shi et al. [3]. ConvLSTM extended traditional LSTM architectures by replacing fully connected operations with convolutional layers, enabling the model to learn both spatial and temporal dependencies simultaneously.

The model demonstrated strong performance in precipitation nowcasting and established the foundation for many subsequent spatiotemporal forecasting architectures. Building upon ConvLSTM, Wang et al. proposed PredRNN [4], which introduced spatiotemporal memory mechanisms to better capture long-term dependencies and improve predictive performance in sequential forecasting tasks.

Despite their success, recurrent forecasting models suffer from several limitations. Forecasts often become increasingly blurry as prediction horizons increase because optimization is typically performed using pixel-wise reconstruction losses. This limitation motivated researchers to explore generative approaches capable of producing sharper and more realistic atmospheric forecasts.

The emergence of diffusion probabilistic models marked a significant advancement in generative artificial intelligence. Ho et al. [1] introduced Denoising Diffusion Probabilistic Models (DDPMs), which learn to generate realistic data through an iterative denoising process. Unlike Generative Adversarial Networks (GANs), diffusion models provide stable training dynamics and superior image quality. Their success in computer vision applications motivated researchers to investigate their applicability in scientific forecasting and meteorological modeling.

A major improvement to diffusion architectures was proposed by Rombach et al. [2] through Latent Diffusion Models (LDMs). Instead of performing diffusion directly in pixel space, LDMs first compress images into a lower-dimensional latent representation using Variational Autoencoders (VAEs). Diffusion is then applied in the latent space, significantly reducing computational cost while preserving high-quality image generation capabilities. This innovation became particularly important for large-scale satellite forecasting systems where computational efficiency is critical.

Recent years have witnessed a rapid increase in the application of diffusion models for weather forecasting. One of the earliest works in this area was LDCast [5], which introduced latent diffusion models for precipitation nowcasting while simultaneously quantifying forecast uncertainty. The study demonstrated that diffusion-based architectures could produce sharper precipitation maps and more reliable uncertainty estimates than conventional deterministic forecasting systems. Similarly, DYffusion [6] proposed a dynamics-informed diffusion framework that incorporated physical constraints into the diffusion process. By integrating atmospheric dynamics into the model architecture, DYffusion improved temporal consistency and forecasting realism.

The potential of diffusion models for ensemble forecasting was demonstrated by SEEDS [7], which showed that diffusion models could emulate computationally expensive weather forecast ensembles while preserving realistic atmospheric variability. This work highlighted the ability of diffusion architectures to model complex probability distributions of atmospheric states, making them suitable for probabilistic weather forecasting. PreDiff [8] further explored latent diffusion for precipitation nowcasting and demonstrated improved cloud morphology

preservation compared to ConvLSTM-based methods.

DiffCast [9] introduced a residual diffusion framework designed specifically for precipitation nowcasting. By modeling residual atmospheric evolution patterns, the model improved forecasting accuracy while maintaining realistic weather structures. Another influential contribution was GenCast [10], which applied diffusion-based ensemble forecasting to medium-range weather prediction. Published in *Nature*, GenCast demonstrated that diffusion models could achieve competitive forecasting performance compared to operational numerical weather prediction systems while requiring significantly fewer computational resources.

Several researchers subsequently focused on improving probabilistic weather forecasting using diffusion architectures. DGDM [11] introduced deterministic guidance mechanisms within the diffusion process to improve forecast stability and uncertainty estimation. CorrDiff [12] utilized residual diffusion techniques for kilometer-scale atmospheric downscaling, enabling the generation of high-resolution weather forecasts from coarse-resolution meteorological inputs. Similarly, SwinRDM [13] combined Swin Transformer architectures with diffusion models to enhance high-resolution weather forecasting and atmospheric detail preservation.

Recent studies have increasingly focused on satellite-based forecasting applications. SATcast [14] proposed a cascade diffusion framework for convective cloud nowcasting using satellite observations. The model demonstrated superior performance in capturing cloud evolution and convective storm development. DiffObs [16] extended diffusion modeling to the global forecasting of satellite observations and highlighted the capability of diffusion architectures to model large-scale atmospheric systems. Likewise, SGD [38] introduced satellite observation-guided diffusion mechanisms that enable meteorological state reconstruction at arbitrary spatial resolutions.

Climate-related applications have also benefited from diffusion-based generative modeling. ClimateDiffuse [15] investigated generative downscaling techniques for climate simulations and demonstrated substantial improvements in preserving regional climate variability. LDM_res [29] explored latent diffusion as a tool for dynamical downscaling of climate simulations, while Latent Diffusion Models for Climate Simulation Ensembles [30] demonstrated the capability of latent diffusion architectures to generate physically consistent climate realizations. Similarly, CPMGEM [31] utilized diffusion models for precipitation emulation from kilometer-scale regional climate simulations.

Data assimilation represents another rapidly growing research area for diffusion-based weather modeling. DiffDA [17] proposed a diffusion-based framework for weather-scale data assimilation, enabling improved integration of observational information into forecasting systems. SLAMS [18] extended this concept by introducing deep generative multimodal data assimilation techniques that incorporate multiple observation modalities. Appa [40] further explored latent diffusion models for global-scale atmospheric data assimilation, demonstrating

promising results in reconstructing realistic atmospheric states.

Extreme weather forecasting has also attracted significant attention. TCP-Diffusion [19] focused on tropical cyclone precipitation forecasting using multimodal diffusion architectures. StormCast [20] proposed convection-allowing weather model emulation using diffusion models and demonstrated improved storm representation compared to deterministic forecasting approaches. FuXi-Extreme [23] and ExtremeCast [24] specifically targeted extreme rainfall and wind prediction, highlighting the ability of diffusion architectures to capture rare but impactful meteorological events.

Researchers have additionally explored uncertainty quantification in diffusion-based weather prediction systems. CoDiCast [21] introduced conditional diffusion forecasting with explicit uncertainty estimation, while ARCI [36] proposed continuous ensemble forecasting through diffusion-based probabilistic modeling. Spherical DYffusion [28] extended diffusion forecasting to global climate modeling on spherical domains, enabling more accurate representation of planetary-scale atmospheric dynamics.

Diffusion models have also been applied to specialized meteorological forecasting tasks. Diffusion Models for High-Resolution Solar Forecasts [22] investigated renewable energy forecasting applications. SDEdit-Weather [25] integrated realistic forecast guidance within the diffusion sampling process to improve atmospheric consistency. Physics-Informed Diffusion Models [26] incorporated physical weather constraints directly into the diffusion process, addressing concerns regarding physical realism and forecast plausibility.

Cloud-related atmospheric forecasting has seen notable progress through diffusion architectures. CloudDiff [27] introduced generative diffusion techniques for cloud property retrieval and super-resolution cloud analysis. DifERS [34] utilized multisensor satellite observations to generate probabilistic precipitation ensembles, demonstrating improved representation of cloud and precipitation uncertainties.

Oceanic and cryospheric forecasting have similarly benefited from diffusion models. KESHDiff [33] applied diffusion techniques to sea surface height downscaling over the Kuroshio Extension, while IceDiff [37] utilized generative diffusion priors for high-resolution sea ice forecasting. These studies demonstrate the versatility of diffusion models across multiple geophysical forecasting domains.

Recent developments have further expanded diffusion forecasting to limited-area and regional weather prediction. Diffusion-LAM [39] introduced probabilistic limited-area weather forecasting using diffusion architectures, while R2-D2 [35] explored dynamical-generative downscaling of climate model ensembles. Forecasting Tropical Cyclones with Cascaded Diffusion Models [32] demonstrated the effectiveness of hierarchical diffusion architectures for tropical cyclone trajectory and intensity prediction.

Collectively, the literature reveals a clear evolution from recurrent forecasting

architectures such as ConvLSTM [3] and PredRNN [4] toward generative diffusion-based weather forecasting systems. Diffusion models consistently demonstrate superior performance in preserving spatial structures, atmospheric textures, cloud morphology, and uncertainty information. Furthermore, latent diffusion architectures significantly reduce computational requirements while maintaining forecasting quality, making them particularly attractive for operational satellite nowcasting systems.

Inspired by these advancements, the proposed Meghdoot-AI framework adopts a latent diffusion approach for satellite weather nowcasting using INSAT-3DR/3DS observations. By leveraging VAE-based latent compression, diffusion-driven atmospheric evolution modeling, and cloud-native deployment infrastructure, Meghdoot-AI aims to provide realistic, scalable, and computationally efficient short-term weather forecasts while addressing limitations observed in traditional ConvLSTM-based systems.

Dataset Description

The Meghdoot-AI framework utilizes geostationary meteorological satellite observations obtained from the Indian National Satellite System (INSAT), specifically INSAT-3DR and INSAT-3DS. These satellites are operated by the Indian Space Research Organisation (ISRO) and provide continuous atmospheric monitoring over the Indian subcontinent and surrounding oceanic regions. The satellite datasets were acquired through the Meteorological and Oceanographic Satellite Data Archival Centre (MOSDAC), which serves as the primary repository for Indian meteorological satellite products.

The acquired data consists of Level-1C calibrated satellite imagery stored in Hierarchical Data Format version 5 (HDF5). Level-1C products provide radiometrically corrected and geolocated atmospheric observations suitable for scientific analysis and machine learning applications. The dataset contains multi-temporal atmospheric imagery capturing cloud evolution, thermal variations, moisture transport, and large-scale atmospheric dynamics.

For this research, approximately 300 GB of satellite data was collected using an automated ingestion pipeline developed in Python. The ingestion system authenticated with the MOSDAC platform, performed bulk downloading of INSAT-3DR/3DS products, validated downloaded files, and synchronized processed outputs with Google Cloud Storage buckets. The automated acquisition workflow enabled large-scale scalable data collection without dependence on manual downloading procedures.

Although INSAT satellites provide multiple atmospheric spectral channels, the current implementation of Meghdoot-AI focuses on two primary atmospheric channels:

Thermal Infrared 1 (TIR1)

Water Vapor (WV)

The TIR1 channel captures cloud-top temperature distributions and thermal atmospheric structures, making it highly suitable for identifying cloud intensity, storm systems, and atmospheric convection patterns. The WV channel captures upper-tropospheric moisture content and atmospheric humidity transport, providing essential information regarding cloud formation and moisture evolution. These two channels were selected because they preserve critical atmospheric information while significantly reducing computational complexity compared to full multi-channel forecasting systems.

The downloaded HDF5 files underwent extensive preprocessing before model training. The preprocessing pipeline was implemented using Python geospatial libraries including Rasterio, GDAL, xarray, h5py, and NumPy. The pipeline performed:

- Atmospheric channel extraction

- Geospatial reprojection

- Indian subcontinent cropping

- Pixel normalization

- Tensor conversion

- Temporal sequence generation

The Indian geographical region was spatially cropped to focus the forecasting system specifically on atmospheric dynamics relevant to the Indian subcontinent. This reduced unnecessary oceanic and non-relevant spatial information while improving training efficiency.

After preprocessing, the atmospheric imagery was converted into PyTorch-compatible tensors suitable for deep learning workflows. Temporal sequences of atmospheric observations were generated to support spatiotemporal forecasting tasks. These processed tensors were subsequently uploaded into Google Cloud Storage buckets for scalable distributed training and cloud-native deployment.

To reduce memory requirements and computational overhead during forecasting, the preprocessed atmospheric tensors were compressed into latent representations using a Variational Autoencoder (VAE). The trained VAE generated approximately 1758 latent atmospheric tensors with dimensions of [4, 64, 64], enabling efficient latent-space forecasting using diffusion models.

The dataset captures diverse atmospheric conditions including:

- Cloud movement

- Cyclonic activity

- Convective cloud evolution

- Thermal atmospheric variations

- Moisture transport patterns

Seasonal atmospheric transitions

This diversity allows the forecasting system to learn complex nonlinear atmospheric dynamics across varying weather conditions.

The final dataset pipeline established a scalable cloud-native atmospheric intelligence workflow integrating automated satellite acquisition, geospatial preprocessing, latent-space generation, and distributed cloud storage infrastructure. The combination of indigenous INSAT satellite observations with generative AI forecasting architectures forms the foundation of the Meghdoot-AI atmospheric nowcasting framework.

Problem Statement

Accurate short-term atmospheric forecasting remains one of the most critical challenges in modern meteorology and disaster management. Traditional Numerical Weather Prediction (NWP) systems require extensive computational resources and often struggle to generate high-resolution localized forecasts in real time. Existing deep learning approaches such as ConvLSTM and recurrent forecasting architectures have improved spatiotemporal weather prediction capabilities; however, they frequently produce blurry outputs, weak cloud boundary preservation, and limited atmospheric texture reconstruction. Furthermore, many existing forecasting systems are not optimized for large-scale cloud-native deployment and often depend on global datasets rather than indigenous satellite infrastructure such as INSAT-3DR/3DS. These limitations reduce their operational effectiveness for real-time atmospheric nowcasting over the Indian subcontinent.

The Meghdoot-AI project aims to address these challenges by developing a cloud-native latent diffusion-based atmospheric nowcasting framework using INSAT-3DR/3DS satellite imagery. The proposed system leverages Thermal Infrared (TIR1) and Water Vapor (WV) channels to learn atmospheric evolution patterns through latent-space forecasting. By integrating Variational Autoencoders (VAE), latent diffusion models, automated satellite ingestion pipelines, geospatial preprocessing, and scalable cloud infrastructure, the framework seeks to generate realistic short-term atmospheric forecasts with improved cloud morphology preservation and spatial continuity. The project also aims to establish an operational end-to-end forecasting pipeline capable of automated data acquisition, GPU-accelerated model training, and scalable deployment for real-time atmospheric intelligence applications.

Implementation

The implementation of Meghdoot-AI was designed as a complete cloud-native atmospheric nowcasting pipeline integrating satellite data acquisition, geospatial preprocessing, latent-space compression, diffusion-based forecasting, cloud storage infrastructure, experiment monitoring, and scalable deployment services. The system architecture was developed to support automated large-scale atmospheric forecasting using INSAT-3DR/3DS satellite observations while ensuring scalability, reproducibility, and operational efficiency.

The implementation pipeline begins with automated satellite data acquisition from the MOSDAC platform. A Python-based ingestion system was developed and executed on Google Colab environments to automate large-scale downloading of INSAT-3DR and INSAT-3DS Level-1C HDF5 satellite products. The ingestion pipeline authenticated with the MOSDAC API, queried available atmospheric products, initiated bulk downloads, validated downloaded files, and handled retry mechanisms for incomplete transfers. Approximately 300 GB of atmospheric satellite data was acquired during the implementation phase. Automating the ingestion process eliminated manual download dependencies and enabled scalable atmospheric dataset collection suitable for deep learning workflows.

After acquisition, the downloaded HDF5 satellite products underwent extensive geospatial preprocessing. The preprocessing pipeline was implemented using Rasterio, GDAL, xarray, h5py, NumPy, and PyTorch-based tensor generation utilities. Although INSAT satellites provide multiple atmospheric spectral channels, the current implementation specifically utilized only two atmospheric channels:

Thermal Infrared 1 (TIR1)

Water Vapor (WV)

The TIR1 channel was selected because it captures cloud-top thermal distributions and atmospheric temperature variations, while the WV channel captures upper-atmospheric moisture transport and humidity patterns. These channels provide critical atmospheric information necessary for cloud evolution forecasting while significantly reducing computational overhead.

The preprocessing workflow performed multiple operations including:

HDF5 atmospheric channel extraction

Geospatial reprojection

Spatial cropping for the Indian subcontinent

Pixel normalization

Tensor conversion

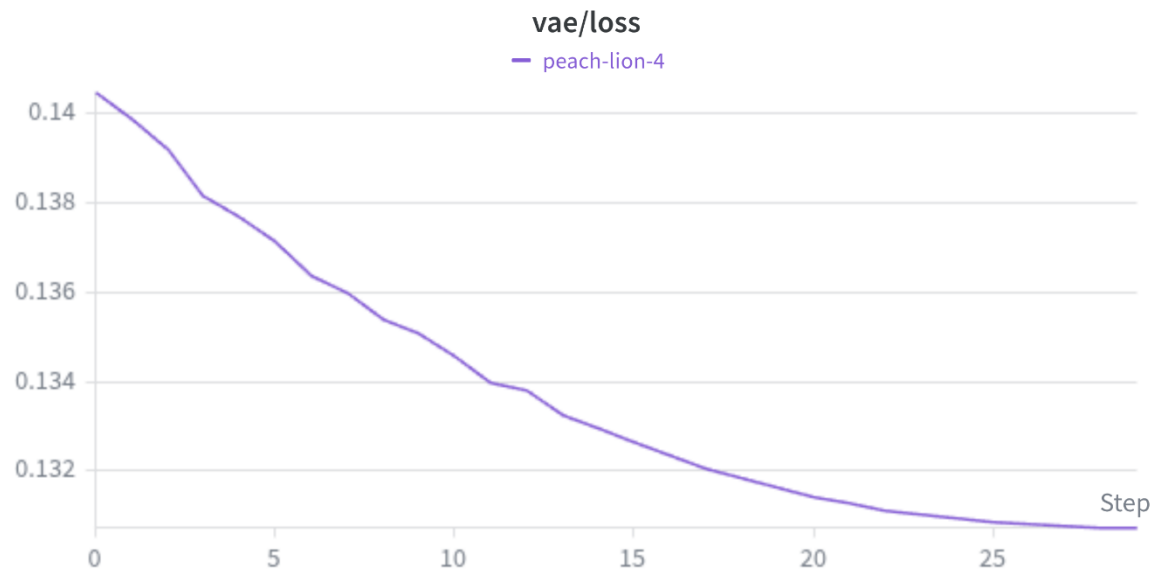
Temporal sequence generation

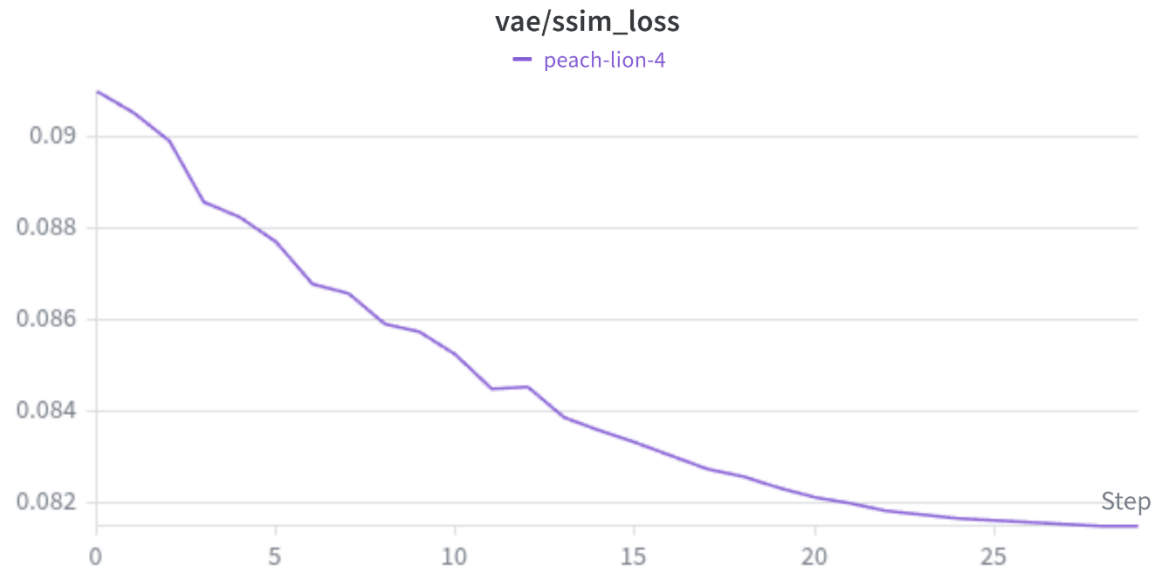
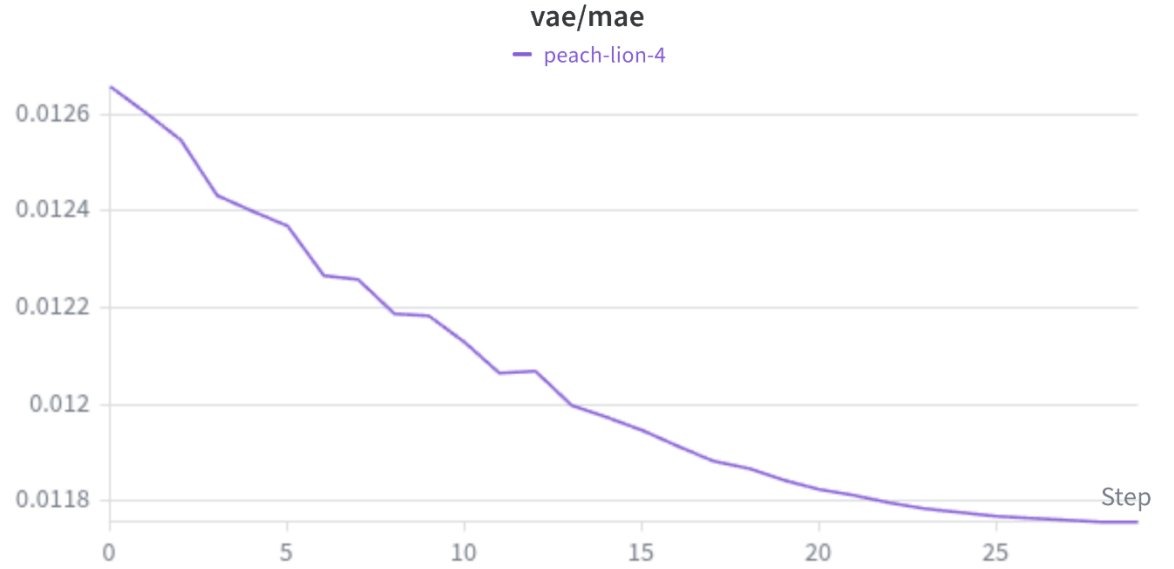
The Indian geographical region was isolated to remove unnecessary spatial information and focus model learning on regional atmospheric dynamics. Pixel values were normalized to stabilize neural network training and improve convergence behavior. The processed atmospheric frames were then converted into PyTorch-compatible tensors for deep learning model development.

To support scalable distributed training, all processed tensors were uploaded and synchronized with Google Cloud Storage (GCS) buckets. The cloud-native storage architecture eliminated local storage bottlenecks and enabled seamless integration with GPU-enabled training pipelines. This design ensured that

preprocessing, training, and inference workflows remained scalable and cloud-accessible across different deployment environments.

A Variational Autoencoder (VAE) was implemented to compress high-dimensional atmospheric imagery into compact latent representations. Direct forecasting on full-resolution satellite imagery requires extremely high memory and computational resources. Therefore, the VAE was designed to encode atmospheric tensors into low-dimensional latent-space representations while preserving critical atmospheric structures such as cloud morphology, thermal gradients, and moisture distributions.





The encoder network progressively reduced spatial dimensions using convolutional layers, generating latent tensors with dimensions of [4, 64, 64]. The decoder reconstructed atmospheric frames from these latent representations. The VAE was trained using reconstruction and KL-divergence losses to balance latent regularization and reconstruction fidelity. Experimental validation demonstrated

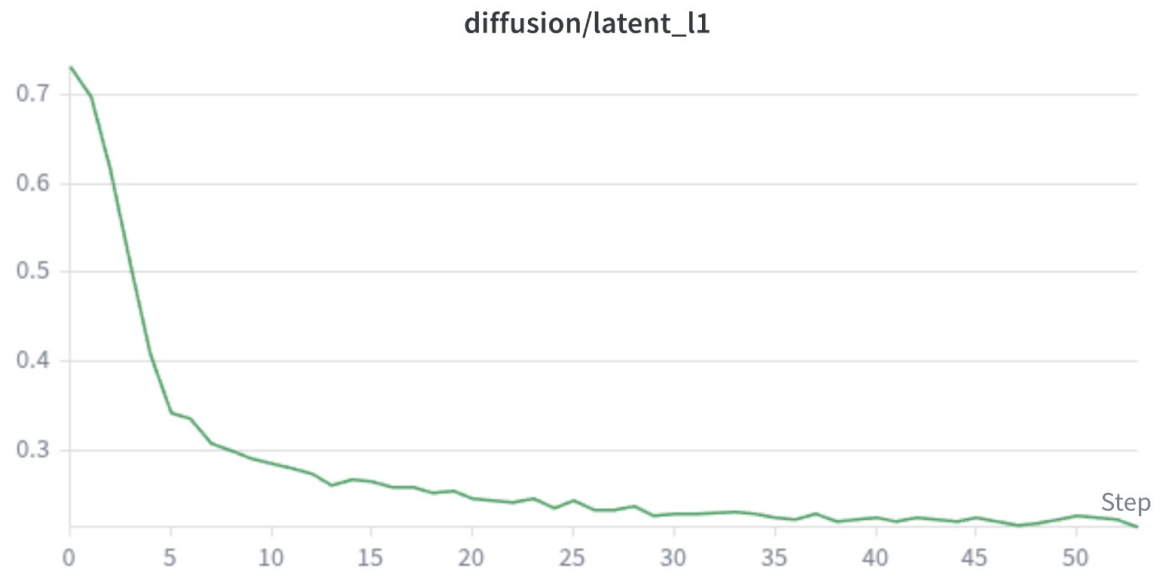
strong reconstruction quality with:

SSIM: 0.8800

PSNR: 35.57

RMSE: 0.0333

After VAE training, latent caching was implemented to accelerate diffusion training. Approximately 1758 latent atmospheric tensors were generated and stored for downstream forecasting workflows. Operating in latent space significantly reduced computational complexity and enabled efficient generative forecasting.



The core forecasting component of Meghdoot-AI was implemented using a latent diffusion model with a U-Net backbone architecture. The diffusion model was designed to learn atmospheric evolution dynamics directly within latent space. Instead of predicting future atmospheric frames deterministically, the diffusion model learns iterative denoising processes capable of reconstructing realistic future atmospheric states.

The latent diffusion training process involved:

Encoding atmospheric frames into latent tensors

Adding controlled Gaussian noise to latent representations

Training the U-Net architecture to iteratively remove noise

Learning temporal atmospheric evolution patterns

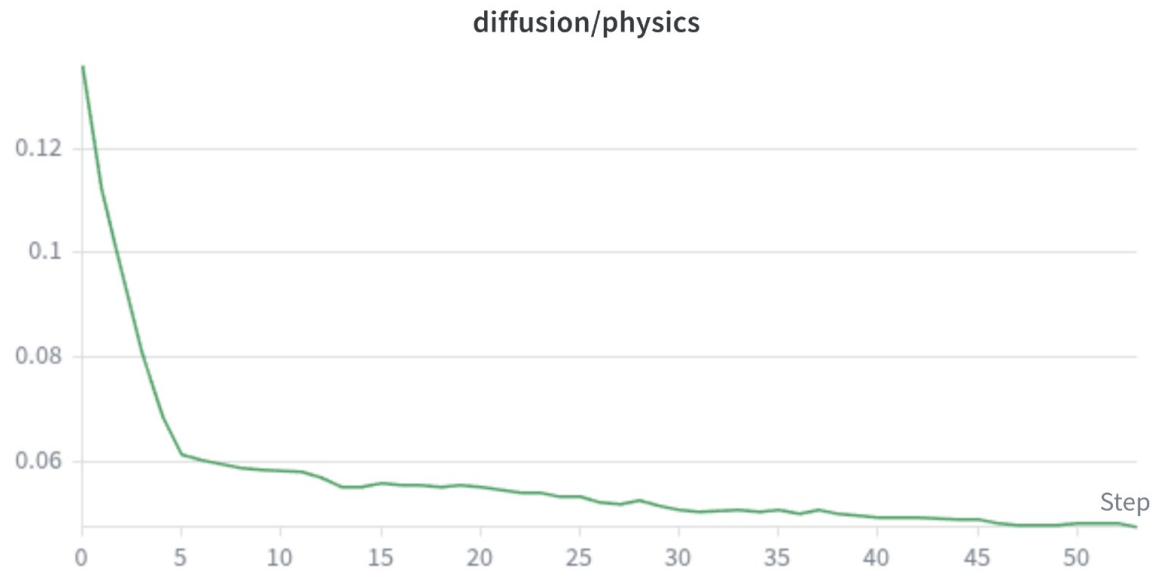
The diffusion framework enabled the system to preserve:

Cloud boundaries

Atmospheric textures

Moisture continuity

Storm morphology



Spatial coherence

Compared to deterministic regression-based forecasting systems, the diffusion architecture generated sharper and more realistic atmospheric forecasts.

To evaluate forecasting performance, the proposed diffusion model was compared against a ConvLSTM baseline architecture. ConvLSTM was selected because it is widely used in spatiotemporal weather forecasting and precipitation nowcasting tasks. Quantitative evaluation was performed using:

Structural Similarity Index (SSIM)

Peak Signal-to-Noise Ratio (PSNR)

Root Mean Square Error (RMSE)

Critical Success Index (CSI)

Experimental results demonstrated that ConvLSTM achieved slightly stronger pixel-level reconstruction metrics, while the latent diffusion framework produced visually sharper atmospheric structures and improved cloud morphology

preservation. The diffusion model also demonstrated competitive CSI scores for atmospheric event detection.

The entire training infrastructure was containerized using Docker to ensure reproducibility and consistent dependency management across development and deployment environments. Docker containers included CUDA-enabled PyTorch environments, geospatial libraries, and forecasting services. GPU acceleration was provided using NVIDIA L4 GPUs, enabling efficient training of the VAE and diffusion architectures.

Experiment monitoring and model tracking were implemented using Weights & Biases (W&B). Training losses, reconstruction quality, generated atmospheric outputs, GPU utilization, and diffusion convergence metrics were continuously logged and visualized through interactive dashboards. This enabled systematic experimentation, debugging, and performance analysis during model development.

For deployment, the system adopted a cloud-native architecture integrating:

- Docker containers

- Google Cloud Build

- Google Cloud Storage

- FastAPI services

- Streamlit visualization dashboards

- Cloud Scheduler

FastAPI services handled model inference and forecast generation, while Streamlit dashboards enabled interactive visualization of atmospheric outputs. Google Cloud Build automated Docker image generation and deployment workflows. Cloud Scheduler enabled periodic automated execution of forecasting pipelines for operational atmospheric nowcasting scenarios.

The final implementation established a complete end-to-end atmospheric intelligence framework capable of:

- Automated satellite data acquisition

- Geospatial preprocessing

- Latent-space atmospheric encoding

- Diffusion-based forecasting

- Quantitative evaluation

- Real-time deployment

- Cloud-native scalability

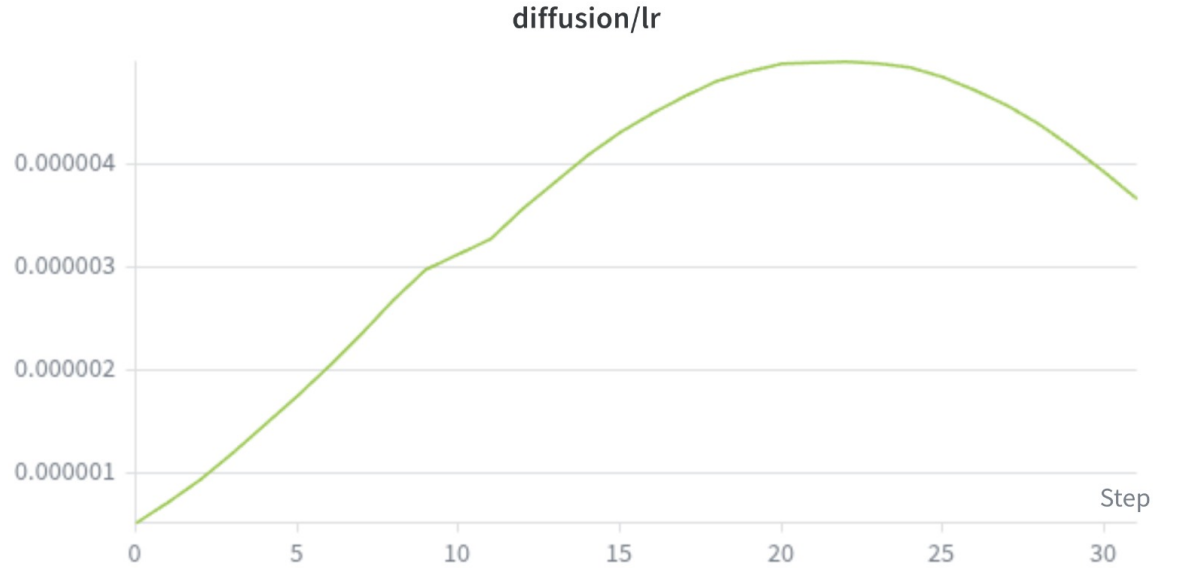
The Meghdoot-AI implementation demonstrates how generative AI, latent diffusion forecasting, geospatial intelligence, and scalable cloud infrastructure can be integrated into a unified operational atmospheric nowcasting system tailored for the Indian subcontinent.

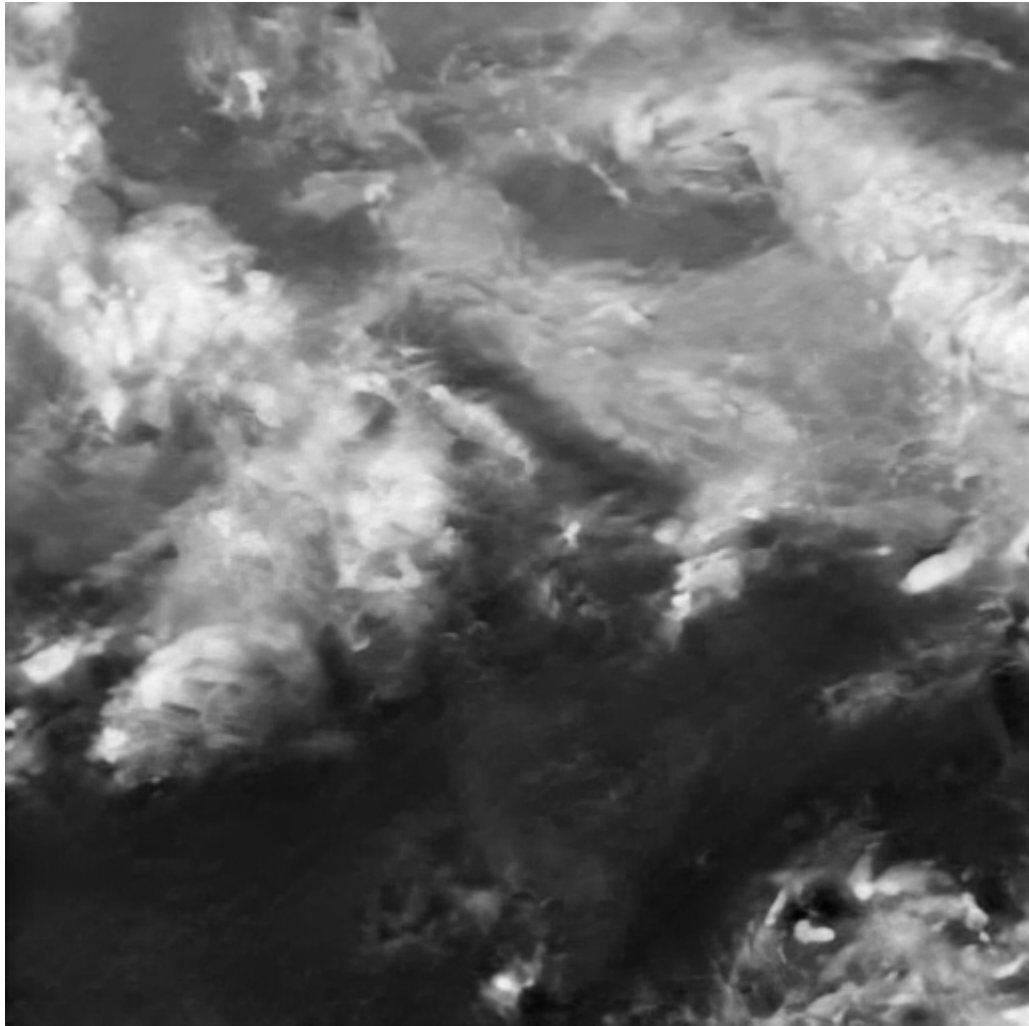
Results and Discussion

The proposed Meghdoot-AI framework was evaluated using INSAT satellite imagery obtained from the MOSDAC platform. The latent diffusion model was trained on preprocessed Thermal Infrared (TIR1) and Water Vapor (WV) channels and compared against a ConvLSTM baseline for short-term satellite nowcasting. Performance evaluation was conducted using both visual inspection and quantitative metrics, including Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), Root Mean Square Error (RMSE), and Critical Success Index (CSI).

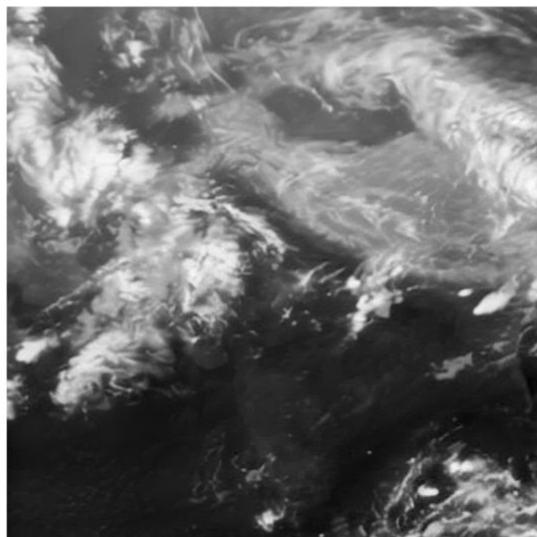
Training Performance

During training, the diffusion model exhibited steady convergence with a continuous reduction in reconstruction and diffusion losses. Visual analysis of intermediate outputs showed progressive improvements in atmospheric reconstruction quality. Early epochs generated relatively blurred cloud structures, whereas later epochs produced sharper cloud boundaries and more realistic atmospheric textures. The model gradually learned complex cloud evolution patterns and successfully captured large-scale atmospheric dynamics.

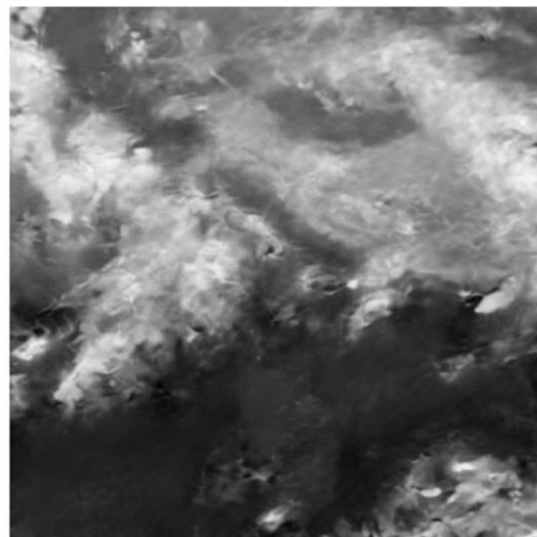




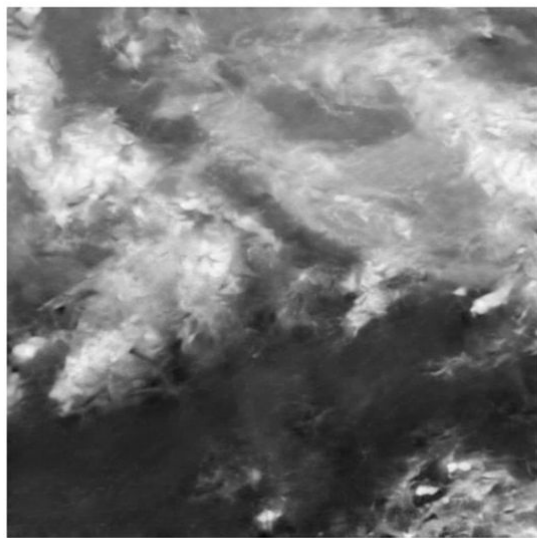
The training progression images demonstrate that forecast quality improved significantly between Epoch 120 and Epoch 180. At later stages, cloud morphology, spatial continuity, and atmospheric texture distributions closely resembled the corresponding ground-truth satellite observations.



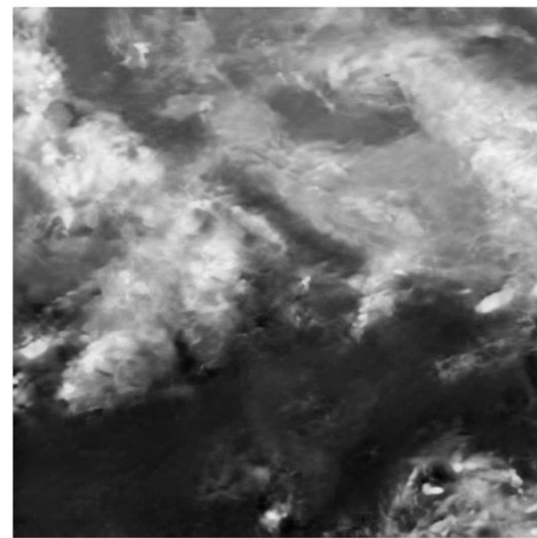
Target



Epoch 120



Epoch 150



Epoch 180

Training Progression

Qualitative Analysis

Visual comparison between the predicted and target satellite images indicates that the latent diffusion model effectively reconstructs large-scale cloud formations and atmospheric structures. The generated forecasts preserve important meteorological features such as cloud clusters, storm patterns, and spatial continuity.

Unlike conventional recurrent architectures that often produce overly smooth predictions, the diffusion model generates sharper outputs with improved texture consistency. The preservation of cloud morphology suggests that the model successfully learns atmospheric evolution dynamics from historical satellite observations. These results highlight the effectiveness of latent diffusion models for realistic satellite nowcasting applications.

Quantitative Evaluation

The forecasting performance of the diffusion model was compared with ConvLSTM using multiple evaluation metrics.

Metric	Diffusion	ConvLSTM
SSIM	0.7260	0.7554
RMSE	0.0822	0.0690
PSNR	28.95	29.52
CSI@600	0.8747	0.8675
CSI@700	0.7859	0.7764
CSI@800	0.5426	0.5980

The results indicate that ConvLSTM achieved slightly higher SSIM and PSNR values while producing lower RMSE. However, the diffusion model demonstrated

superior performance at CSI@600 and CSI@700 thresholds, indicating stronger detection capability for medium-intensity weather events. This suggests that the diffusion model is particularly effective at preserving storm-related atmospheric structures and event-focused forecasting characteristics.

Although ConvLSTM achieved better performance at the highest threshold (CSI@800), the diffusion model maintained competitive accuracy while generating visually more realistic forecasts. The ability of diffusion models to preserve atmospheric textures and cloud evolution patterns provides an important advantage for operational nowcasting applications.

Cloud Deployment Results

The complete forecasting pipeline was successfully deployed on Google Cloud Platform using Docker, Cloud Run, Cloud Build, Artifact Registry, and Cloud Scheduler. Automated forecasting jobs were executed periodically through Cloud Scheduler, enabling real-time generation of satellite nowcasts. The deployed web interface provided visualization of forecast outputs and performance metrics, demonstrating the feasibility of deploying diffusion-based weather forecasting systems in production environments.

Discussion

The experimental results demonstrate that latent diffusion models represent a promising alternative to traditional recurrent forecasting architectures. While ConvLSTM achieves marginally better pixel-level similarity metrics, the diffusion model generates forecasts with improved visual realism and stronger event-detection performance at operational thresholds. The integration of latent-space learning through a Variational Autoencoder significantly reduces computational requirements while preserving important atmospheric features.

Overall, the proposed Meghdoot-AI framework successfully combines latent diffusion forecasting with cloud-native deployment infrastructure, providing an efficient and scalable solution for satellite weather nowcasting. The results validate the effectiveness of diffusion-based approaches for modeling complex atmospheric dynamics and generating realistic short-term weather forecasts.

System Architecture

The architecture of Meghdoot-AI was designed as a scalable cloud-native atmospheric nowcasting framework integrating automated satellite ingestion, geospatial preprocessing, latent-space compression, diffusion-based forecasting, cloud storage infrastructure, experiment tracking, and real-time deployment services. The system follows a modular pipeline-based architecture where each component performs a dedicated atmospheric intelligence task while remaining scalable and loosely coupled.

The overall architecture can be divided into six major layers:

Data Acquisition Layer

Geospatial Preprocessing Layer

Latent Compression Layer

Diffusion Forecasting Layer

Cloud Infrastructure and Deployment Layer

Visualization and Monitoring Layer

The first layer of the architecture consists of the automated satellite data acquisition pipeline. INSAT-3DR and INSAT-3DS Level-1C satellite imagery was collected through the MOSDAC platform using Python-based automated ingestion scripts executed on Google Colab. The ingestion system authenticated with MOSDAC APIs, queried atmospheric products, downloaded HDF5 satellite files in bulk, validated file integrity, and transferred the processed outputs to Google Cloud Storage buckets. This automation eliminated manual data collection dependencies and enabled scalable acquisition of approximately 300 GB of atmospheric observations.

The second layer performs geospatial preprocessing and atmospheric tensor generation. The raw INSAT HDF5 files cannot be directly used for machine learning because they contain multi-dimensional geospatial atmospheric observations requiring normalization and spatial processing. The preprocessing subsystem was implemented using Rasterio, GDAL, xarray, h5py, and NumPy. During preprocessing, only two atmospheric channels were retained:

Thermal Infrared 1 (TIR1)

Water Vapor (WV)

These channels were selected because they preserve critical atmospheric information such as cloud-top thermal structures and moisture transport dynamics while minimizing computational overhead. The preprocessing pipeline performed:

HDF5 parsing

Atmospheric channel extraction

Spatial cropping for the Indian region

Geospatial reprojection

Pixel normalization

Tensor generation

Temporal sequence preparation

The generated tensors were subsequently stored in Google Cloud Storage to support distributed cloud-based training.

The third architectural layer consists of the Variational Autoencoder (VAE)-based latent compression module. Forecasting directly on high-resolution atmospheric imagery is computationally expensive and memory intensive. Therefore, a

VAE was implemented to compress atmospheric tensors into compact latent-space representations. The encoder network progressively reduced the spatial dimensionality of atmospheric frames while preserving essential cloud structures and atmospheric dynamics. The decoder reconstructed atmospheric frames from latent representations during training.

The VAE transformed atmospheric imagery into latent tensors of dimension:

[4, 64, 64]

This latent-space representation significantly reduced memory consumption and enabled efficient forecasting workflows. After VAE training, latent caching was implemented to generate approximately 1758 latent atmospheric tensors for downstream diffusion training.

The fourth and most critical layer of the architecture is the latent diffusion forecasting module. The forecasting engine was implemented using a U-Net-based latent diffusion architecture. Instead of predicting atmospheric frames directly, the diffusion model learns atmospheric evolution patterns in latent space using iterative denoising processes.

The diffusion forecasting workflow operates as follows:

Historical atmospheric frames are encoded into latent tensors.

Controlled Gaussian noise is progressively added during training.

The U-Net diffusion model learns to remove noise iteratively.

The model reconstructs future atmospheric latent representations.

The VAE decoder converts predicted latent tensors back into visible atmospheric imagery.

This generative forecasting mechanism enables realistic reconstruction of:

Cloud morphology

Atmospheric textures

Thermal continuity

Moisture structures

Spatial atmospheric dynamics

Compared to deterministic forecasting systems, the diffusion model produces sharper and more physically consistent atmospheric predictions.

The fifth architectural layer consists of cloud infrastructure and deployment services. The complete Meghdoot-AI framework was designed as a cloud-native system integrating:

Google Cloud Storage (GCS)

Docker containers

FastAPI inference services

Google Cloud Build

Cloud Scheduler

GPU-enabled training environments

All forecasting services were containerized using Docker to ensure reproducibility and dependency consistency across development and deployment environments. NVIDIA L4 GPUs were utilized for accelerated VAE and diffusion model training.

Google Cloud Build automated container image generation and deployment workflows. Cloud Scheduler periodically triggered atmospheric nowcasting pipelines for operational inference execution. The use of cloud-native infrastructure enabled scalable distributed forecasting workflows while eliminating local infrastructure limitations.

The sixth architectural layer handles visualization, experiment tracking, and monitoring. A Streamlit-based dashboard was developed to visualize:

Ground truth atmospheric frames

Diffusion-generated forecasts

ConvLSTM outputs

Evaluation metrics

Forecast lead times

This interface enables interactive atmospheric analysis and qualitative comparison of forecasting outputs.

Experiment monitoring and model tracking were implemented using Weights & Biases (W&B). The monitoring subsystem tracked:

Diffusion training loss

VAE reconstruction quality

GPU utilization

Forecast outputs

Training convergence

Experimental configurations

This enabled systematic experiment management and reproducibility during model development.

The complete Meghdoot-AI architecture establishes an end-to-end operational atmospheric intelligence system integrating:

Automated satellite ingestion

Geospatial preprocessing

Latent-space atmospheric encoding

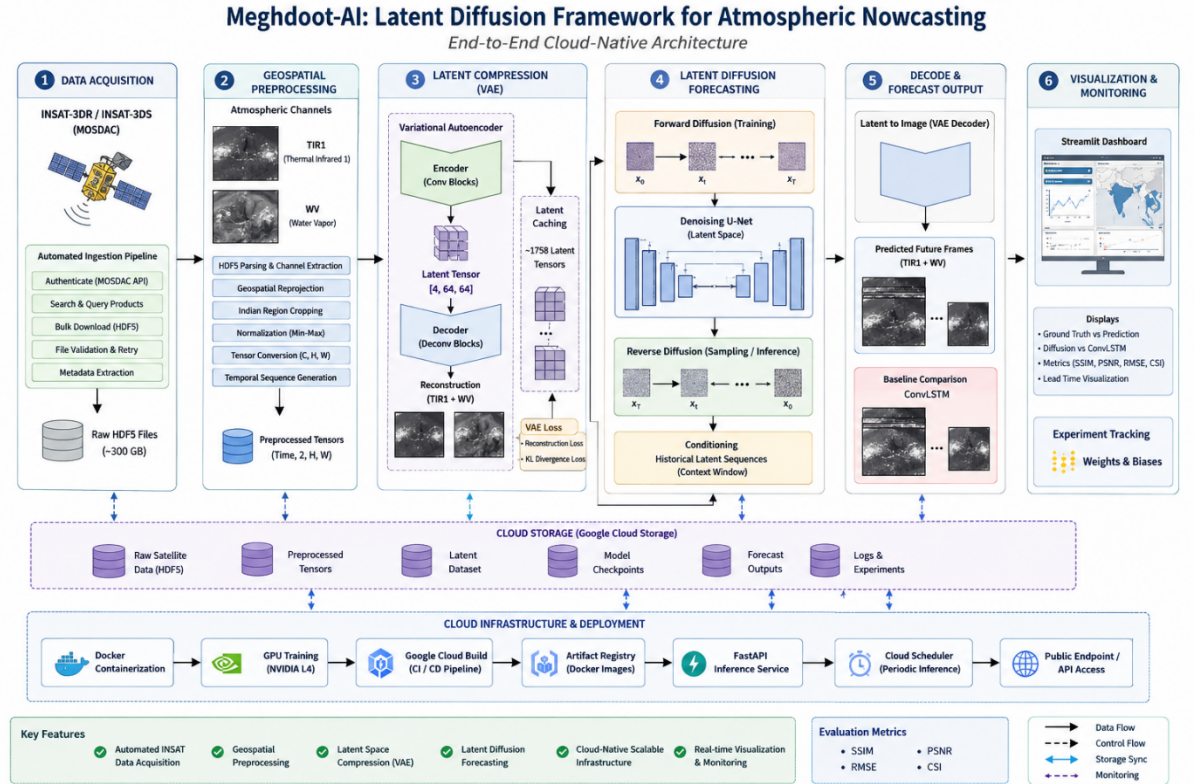
Diffusion-based forecasting

Cloud-native deployment

Real-time visualization

Experiment monitoring

The architecture demonstrates how generative AI, geospatial intelligence, cloud computing, and scalable MLOps infrastructure can be combined into a unified atmospheric nowcasting framework for real-time weather forecasting over the Indian subcontinent.



Comparative analysis

Comparative analysis was performed to evaluate the effectiveness of the proposed Meghdoot-AI latent diffusion framework against the ConvLSTM baseline architecture for atmospheric nowcasting tasks. ConvLSTM was selected as the baseline model because it is one of the most widely adopted deep learning architectures for spatiotemporal weather forecasting and precipitation nowcasting.

The comparison focused on evaluating both quantitative forecasting accuracy and qualitative atmospheric reconstruction quality using INSAT-3DR/3DS satellite observations.

Both models were trained using identical atmospheric datasets consisting of Thermal Infrared 1 (TIR1) and Water Vapor (WV) channels extracted from INSAT satellite imagery. The dataset underwent identical preprocessing operations including spatial cropping, normalization, tensor generation, and temporal sequence preparation to ensure fair experimental comparison. Quantitative evaluation was conducted using multiple atmospheric forecasting metrics including Structural Similarity Index (SSIM), Root Mean Square Error (RMSE), Peak Signal-to-Noise Ratio (PSNR), and Critical Success Index (CSI).

Experimental results demonstrated that the ConvLSTM model achieved slightly stronger pixel-level reconstruction metrics. ConvLSTM produced an SSIM score of 0.7554 compared to 0.7260 achieved by the diffusion model. Similarly, ConvLSTM achieved lower RMSE values and marginally higher PSNR values, indicating improved pixel-wise reconstruction consistency. These results are expected because ConvLSTM architectures optimize deterministic regression objectives, which often minimize pixel-level forecasting errors.

However, qualitative evaluation revealed several important advantages of the latent diffusion forecasting framework. The diffusion model generated significantly sharper atmospheric structures and preserved cloud morphology more effectively than ConvLSTM outputs. ConvLSTM predictions frequently appeared smoother and exhibited reduced texture fidelity due to averaging effects commonly observed in regression-based forecasting systems. In contrast, the diffusion architecture preserved high-frequency atmospheric textures, cloud boundaries, moisture continuity, and mesoscale atmospheric structures with greater realism.

Critical Success Index (CSI) analysis further demonstrated competitive forecasting capabilities for the diffusion framework. The latent diffusion model achieved stronger CSI scores at medium atmospheric thresholds, particularly CSI@600 and CSI@700, indicating improved atmospheric event detection and cloud structure preservation in several forecasting scenarios. Although ConvLSTM achieved stronger CSI performance at higher thresholds, the diffusion model demonstrated more realistic atmospheric evolution behavior during qualitative evaluation.

Another important distinction between the two architectures lies in their forecasting methodology. ConvLSTM performs deterministic frame prediction directly in pixel space using recurrent spatiotemporal operations. While computationally efficient, this approach often struggles to preserve fine-grained atmospheric variability and realistic cloud evolution patterns over longer forecasting horizons. The Meghdoot-AI framework, on the other hand, performs forecasting within compressed latent space using iterative denoising processes. This generative forecasting mechanism enables the diffusion architecture to reconstruct physically consistent atmospheric states while preserving structural atmospheric continuity.

From an architectural perspective, the proposed latent diffusion framework also

demonstrated improved scalability for cloud-native deployment workflows. The integration of Variational Autoencoder-based latent compression significantly reduced computational complexity compared to full-resolution pixel-space forecasting systems. Operating within latent space reduced memory requirements and enabled efficient GPU-accelerated training using NVIDIA L4 infrastructure.

The comparative analysis demonstrates that although ConvLSTM remains highly effective for deterministic pixel-level forecasting tasks, latent diffusion models offer substantial advantages for realistic atmospheric generation, cloud morphology preservation, and generative weather forecasting applications. The ability of diffusion models to preserve atmospheric textures and realistic cloud evolution patterns makes them particularly promising for next-generation atmospheric nowcasting systems.

Overall, the experimental comparison validates the effectiveness of the Meghdoot-AI framework as a scalable generative forecasting system capable of producing realistic atmospheric predictions using INSAT satellite observations. The results indicate that latent diffusion forecasting represents a promising direction for future cloud-native weather intelligence and atmospheric nowcasting research.

Conclusion

The Meghdoot-AI framework presents a cloud-native generative atmospheric nowcasting system designed using latent diffusion models and INSAT-3DR/3DS satellite imagery. The proposed architecture integrates automated satellite data acquisition, geospatial preprocessing, latent-space compression, diffusion-based forecasting, cloud-native deployment infrastructure, and real-time visualization into a unified operational forecasting pipeline. By leveraging Thermal Infrared (TIR1) and Water Vapor (WV) channels, the framework successfully captures critical atmospheric dynamics required for short-term weather forecasting over the Indian subcontinent.

The implementation demonstrated the successful acquisition and processing of more than 300 GB of atmospheric satellite observations through automated MOSDAC ingestion pipelines executed on Google Colab and synchronized with Google Cloud Storage infrastructure. Geospatial preprocessing workflows using Rasterio, GDAL, xarray, and PyTorch enabled scalable atmospheric tensor generation suitable for deep learning applications. The Variational Autoencoder (VAE) effectively compressed atmospheric imagery into latent representations while preserving essential cloud morphology and atmospheric structures. Experimental evaluation achieved strong reconstruction performance with SSIM values of 0.8800, PSNR of 35.57, and low RMSE values, validating the effectiveness of latent-space atmospheric encoding.

The latent diffusion forecasting framework demonstrated promising atmospheric nowcasting capabilities through realistic cloud evolution generation and preservation of atmospheric textures. Comparative analysis against ConvLSTM baseline architectures revealed that while ConvLSTM achieved slightly stronger pixel-level reconstruction metrics, the diffusion model generated sharper atmospheric struc-

tures, improved cloud morphology preservation, and more realistic atmospheric continuity. These findings highlight the advantages of generative diffusion-based forecasting approaches for next-generation atmospheric intelligence systems.

Another major contribution of this research is the integration of scalable cloud-native infrastructure and MLOps workflows into atmospheric forecasting. Dockerized GPU-enabled environments, FastAPI inference services, Google Cloud Build pipelines, Cloud Scheduler automation, and Weights & Biases experiment tracking collectively enabled reproducible, scalable, and operational deployment workflows. The framework demonstrates how geospatial AI, generative modeling, cloud computing, and modern deployment architectures can be combined into an end-to-end atmospheric intelligence platform.

Overall, Meghdoot-AI establishes a strong foundation for real-time atmospheric nowcasting using latent diffusion models and indigenous satellite infrastructure. The proposed system demonstrates the potential of generative AI for weather forecasting applications while contributing toward scalable cloud-native atmospheric intelligence systems for disaster management, climate monitoring, and operational forecasting applications.

Future work will focus on extending the framework toward multi-channel atmospheric forecasting, probabilistic ensemble generation, transformer-assisted diffusion architectures, longer lead-time forecasting horizons, and integration of additional meteorological datasets including radar and reanalysis observations. Further optimization of inference efficiency and operational deployment workflows will also be explored to support large-scale real-time forecasting applications.

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